Computer Science & IT

C Programming



Functions & storage classes

Lecture No. 03



By- Abhishek Sir

Recap of Previous Lecture



Topic — Function , prototype , parameter & arguments

Topic — Function Declaration

Topic — Function definition

Topic — Function call

Topic

Slide

Topics to be Covered



- Topic — Activation tree
- Topic — Storage class
- Topic
- Topic
- Topic

Slide



#Q

GATE 2005

```
double foo (double);                /* Line 1 */
int main() {
    double da, db;
    //input da
    db = foo(da);
}
double foo (double a) {
    return a;
}
```

The above code compiled without any error or warning. If Line 1 is deleted, the above code will show:

A. no compile warning or error

B. some compiler-warnings not leading to unintended results

C. some compiler-warnings due to type-mismatch eventually leading to unintended results

D. compiler errors

Slide

[D] ANSI C - Cgg group discussion

[C] [D] flow

Error

gcc - Linux

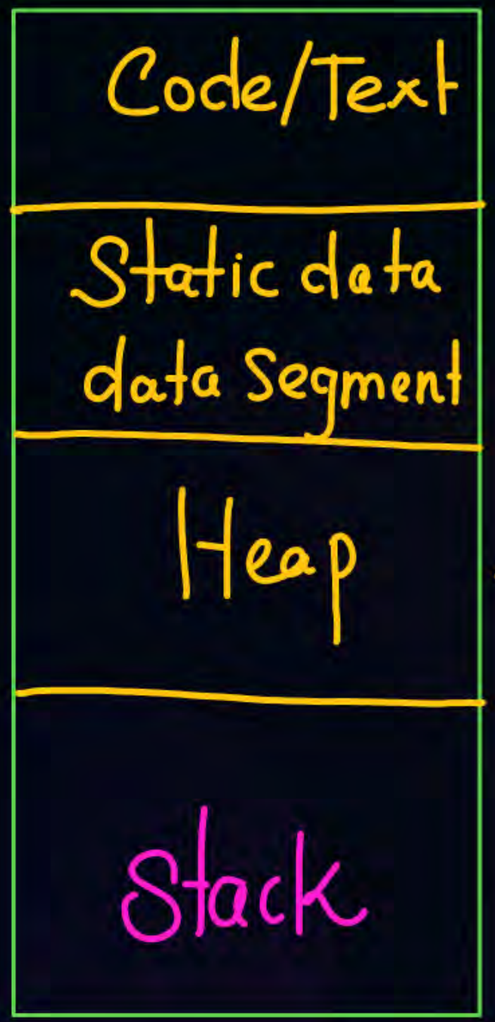
Warning / Error

Cgg

No error

Error

Activation Tree and Records



Exe file ← Load Here for execution by system program called Load

global variable (~static) → storage class

* Dynamic Allocation of Memory
malloc() / calloc()

Main Memory Suppose

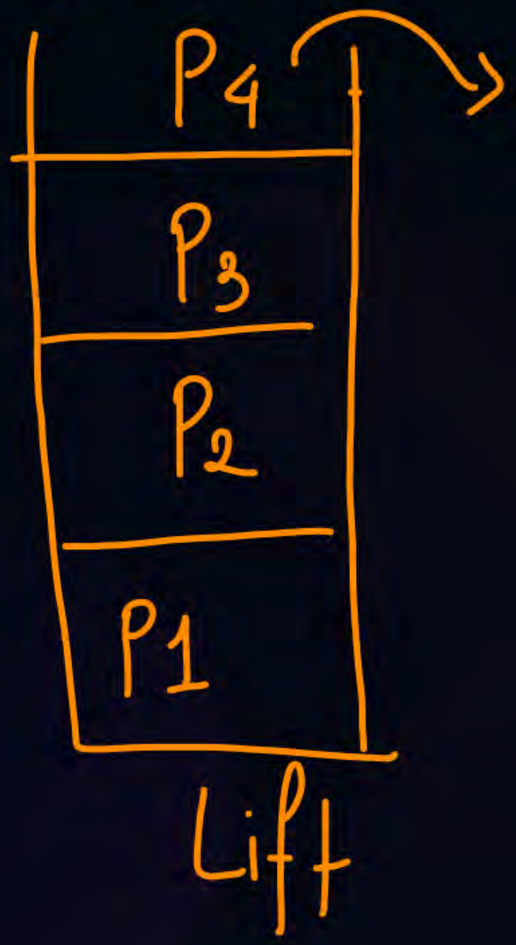
Slide

Activation Tree and Records

Stack :: Data structure (Collection of Data

How to manage)

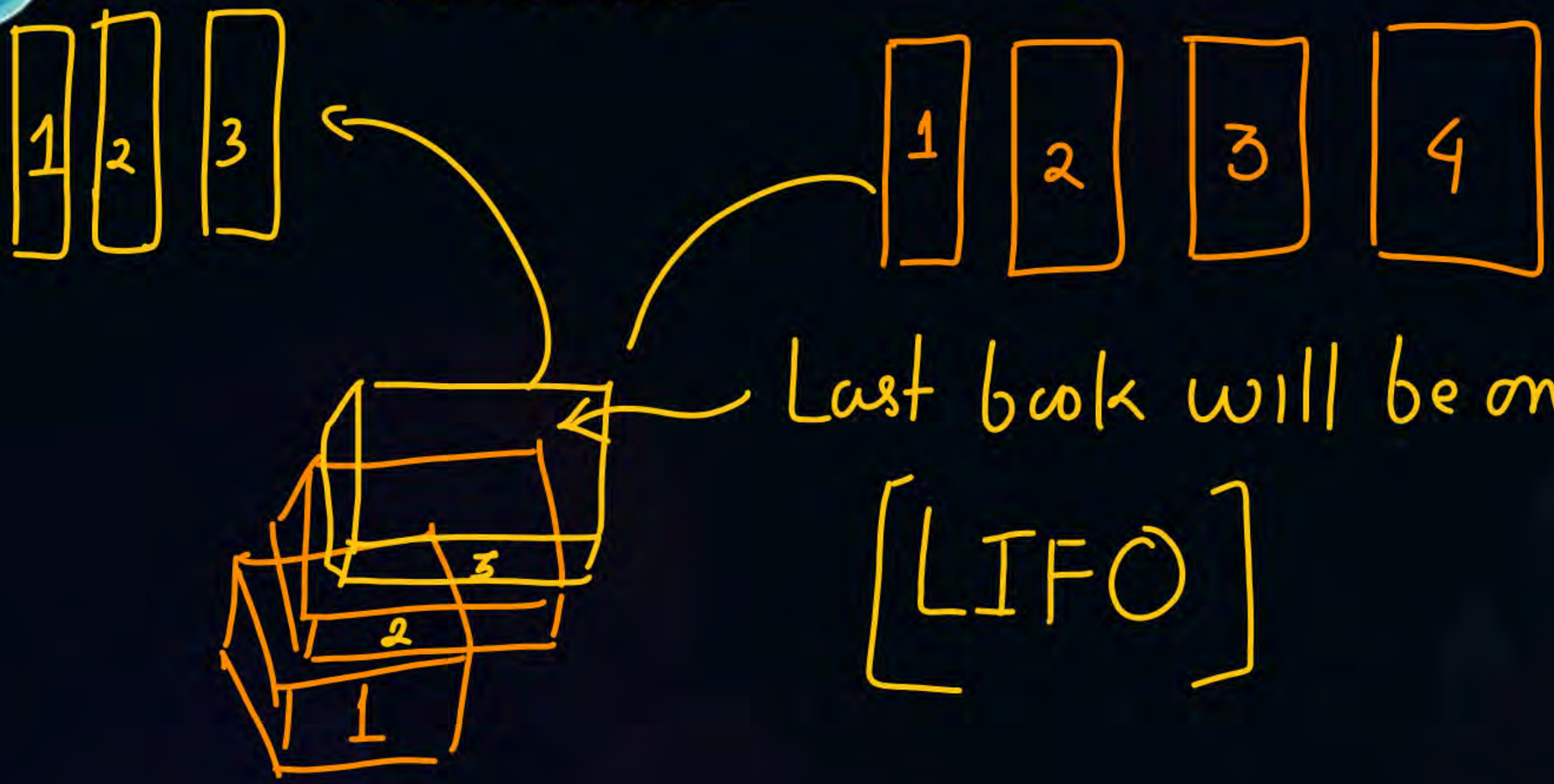
Last in first out (LIFO)



Slide



Activation Tree and Records



Slide



Activation Tree and Records

Activation Record : The information regarding execution of a function is maintained in Record (file) format called Activation Record.

Local variable stored in Activation Record

Slide



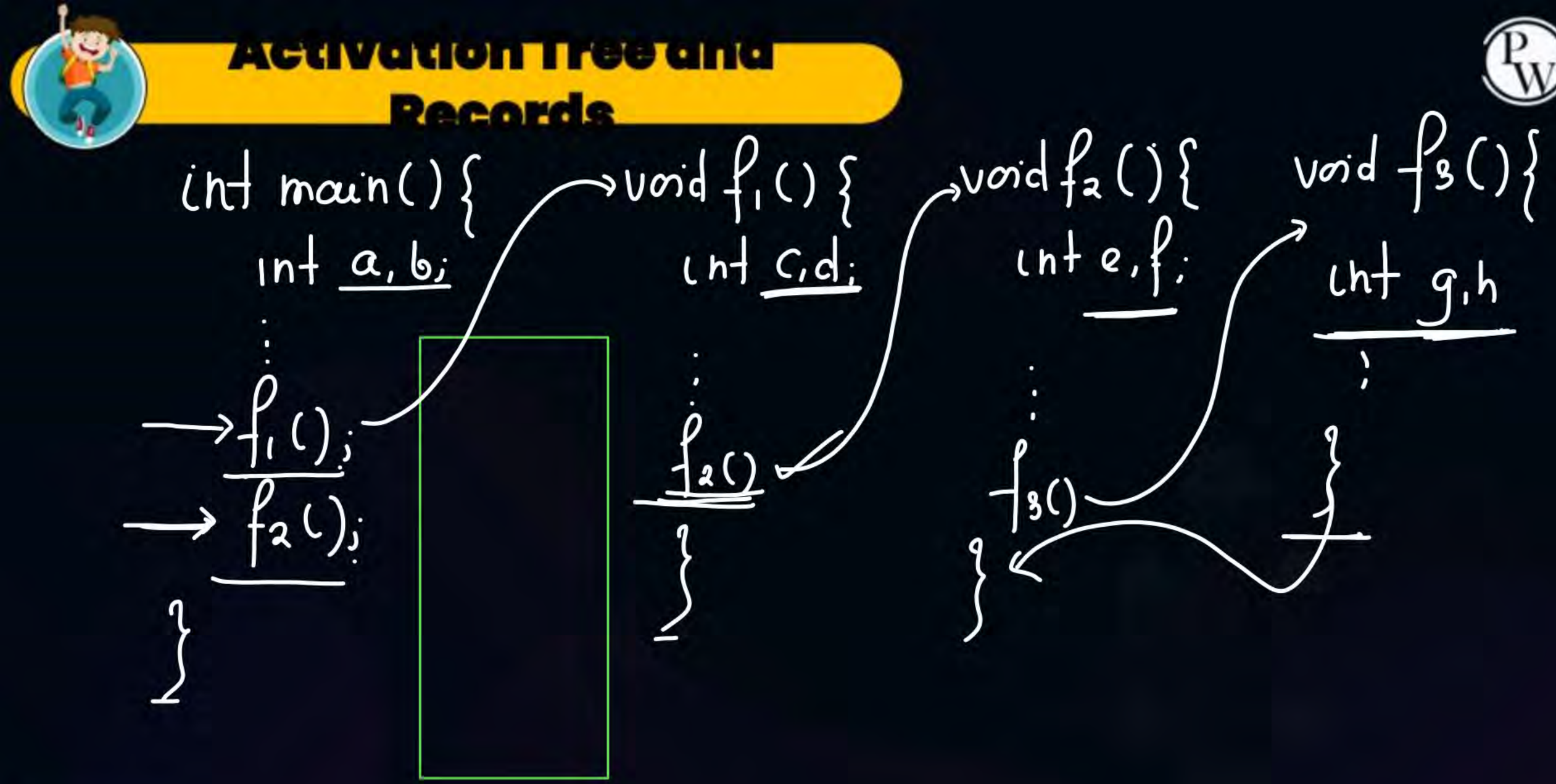
Activation Tree and Records

Local variable declared stored in Stack in form of Activation Record.

(Run time stack)

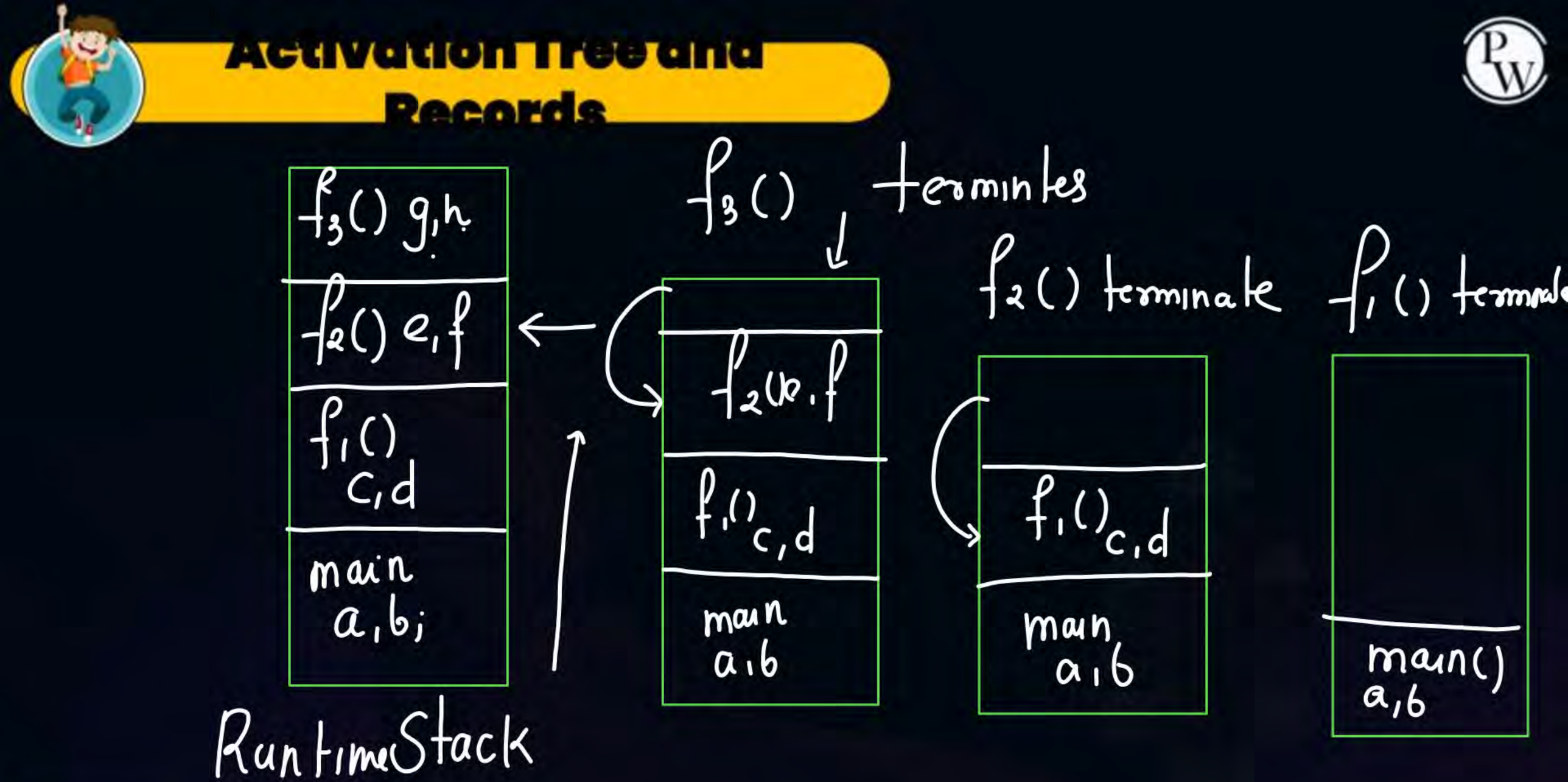
Slide

Activation Tree and Records



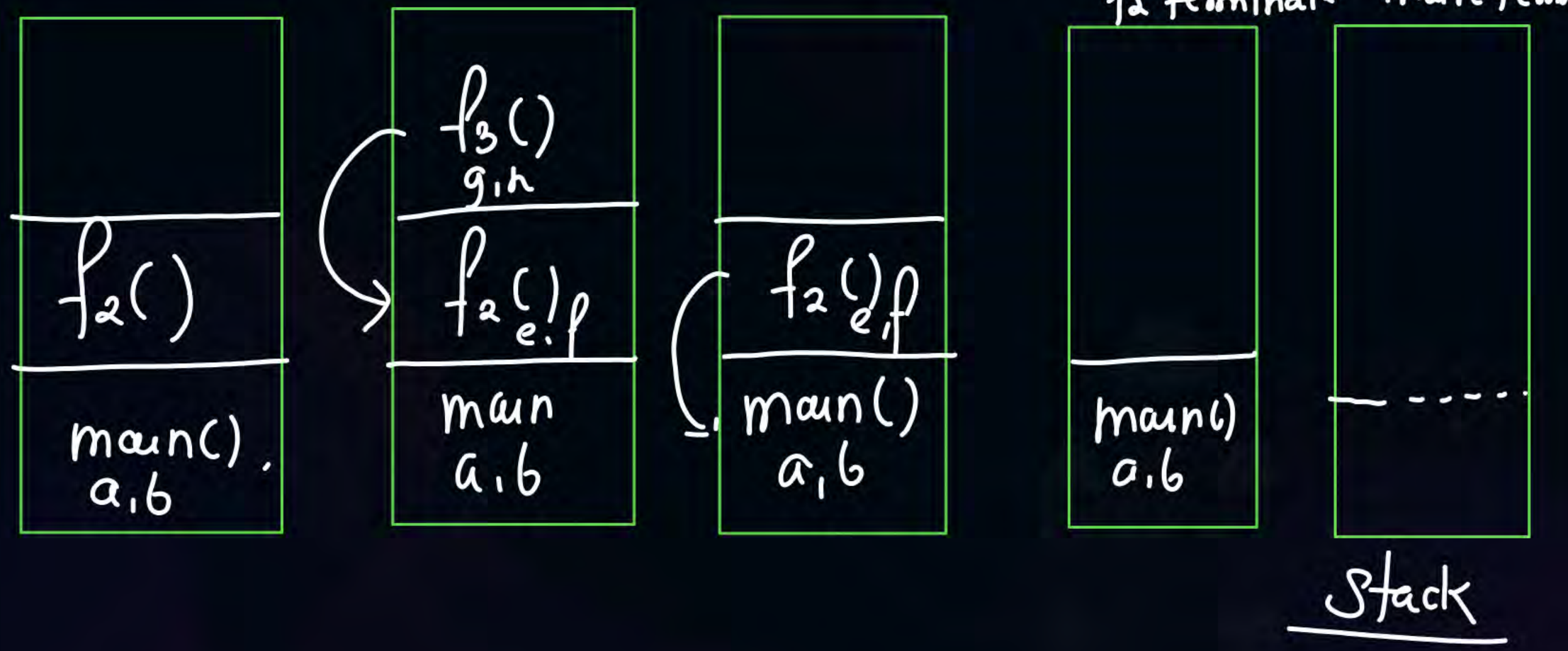
Slide

Activation Tree and Records



Slide

Storage Class



Slide



Storage Class

Second time when f_2 executed will it start with old value or New value.

When Activation Record goes out of stack values are destroyed. (deallocation)

Second time it will start with new assignment.



Storage Class

```
#include<stdio.h>
int f1(int);
int main(){
    int a=50;
    a = f1(a);
    printf("%d", a);
    a = f1(a);
    printf("%d", a);
}
```

70,90

```
int f1(int c){
    return c+20;
}
```

return 0, fut call

C=50

second call C=70

Call by value



Storage Class

```
#include<stdio.h>
int f1(int);
int main(){
    int a=50;
    printf("%d", f1(a));
    printf("%d", f1(a));
    return 0;
}
```

50

61

61

Stack

11

```
int f1(int c){
    int d=10;
    d++; // ++d;
    return c+d;
}
```

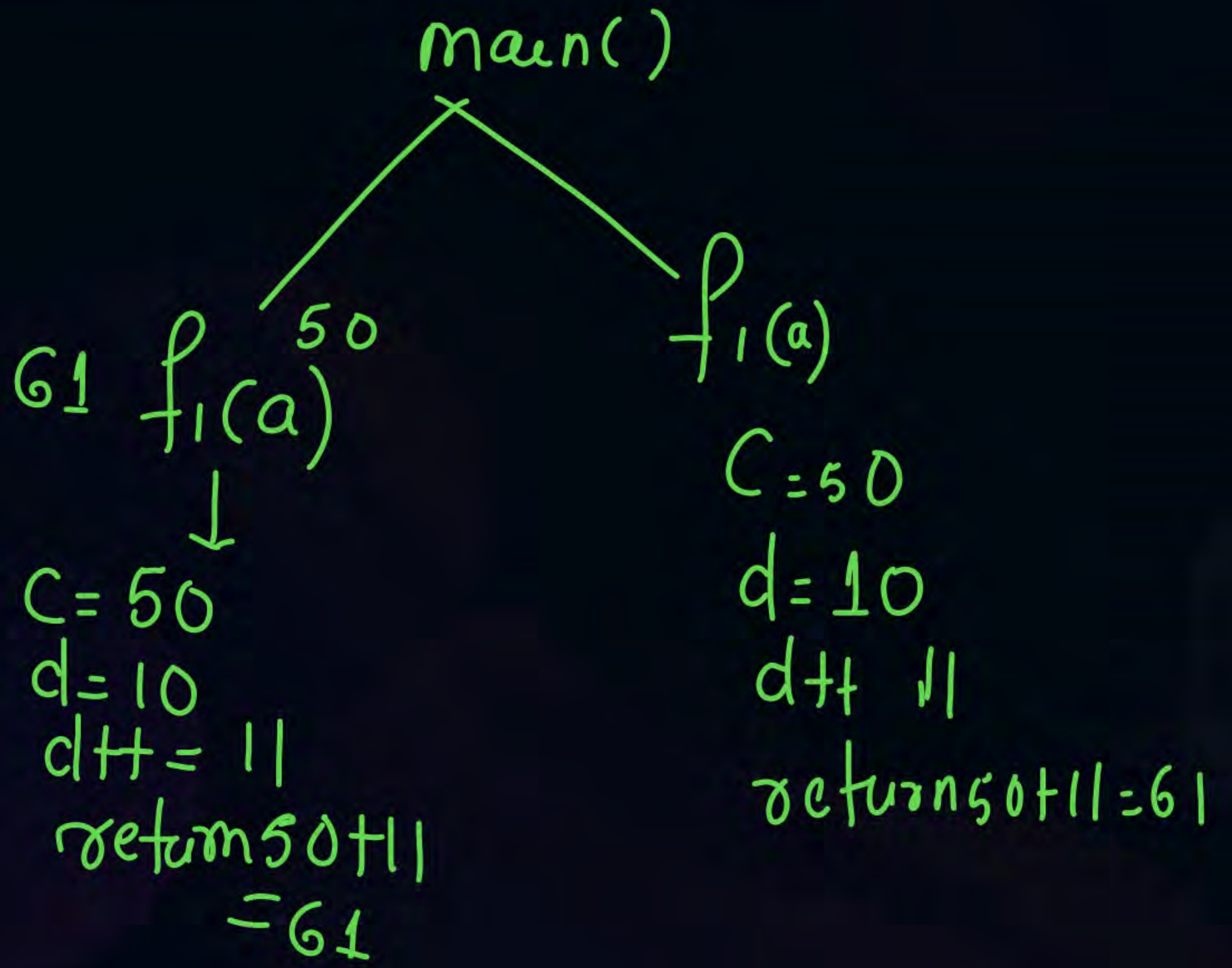
50+11=61

61	61

Slide

Storage Class





Slide



Storage Class

1. auto
2. Static
3. Extern
4. Register

The defaut storage class for Local variable is auto;

```
int f1 (int c){
auto int d;
:
}
```

Slide



Storage Class

Local variable allocated in stack and when function terminates local variable destroyed

auto int a;

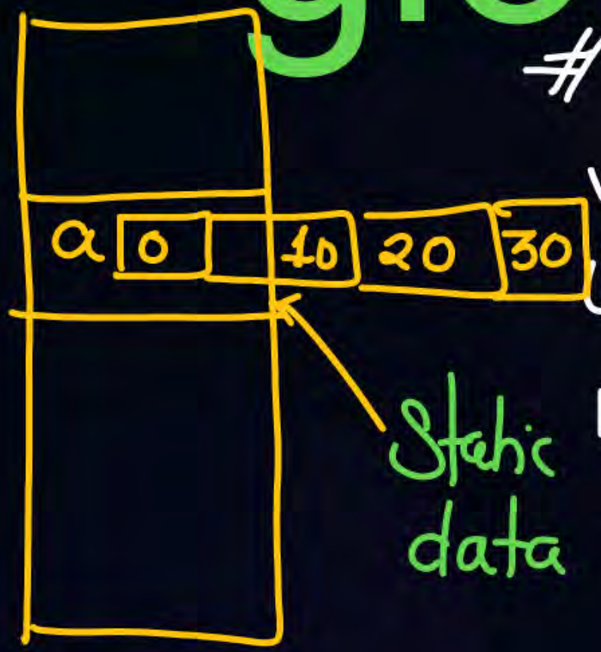
auto

Stack

Slide

Storage Class

global variable




```
# include<stdio.h>
void f1()
int a;
int main() {
  f1();
  f1();
  f1();
```

30 ←

```
printf("%d", a);
}
```

```
void f1() {
  a=a+10;
}
```

Slide



Storage Class

If local variable not initialized then it contains garbage value.

global variable automatically initialized to '0';

The default storage class for global variable is **static**



Storage Class

* Static variable automatically ~~init~~ initialized to '0'
* global variable visible to every function defined in program
* Single memory allocated in static data / data segment
* if static variable is local then value initialized only once.

Slide



Storage Class

C | 20 40 | 60 ← data segment / static

Local variable can also declared as static

```
# include<stdio.h>
void f1(int);
int main(){
    int a=20;
    f1(a);
    f1(a);
    return 0;
}
```

20

20

40,60

```
void f1(int b){
    Static int C=20;
    C=C+b;
    printf("%d", C);
}
```

Second time will not execute

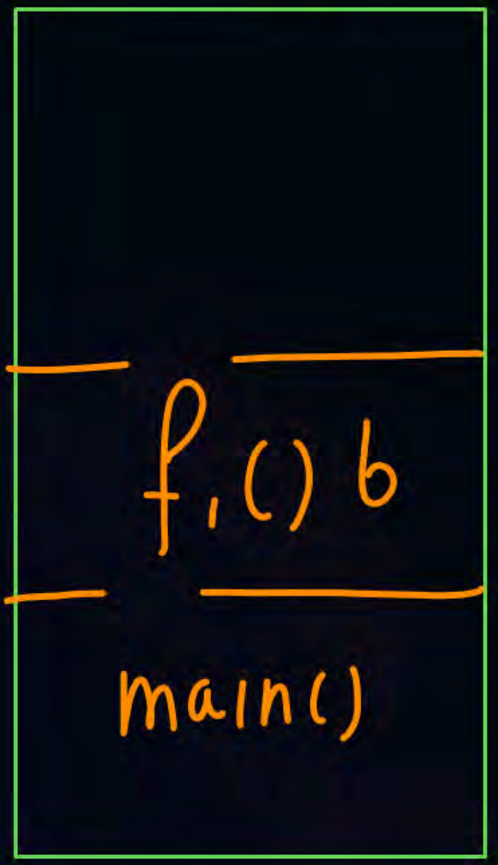
40

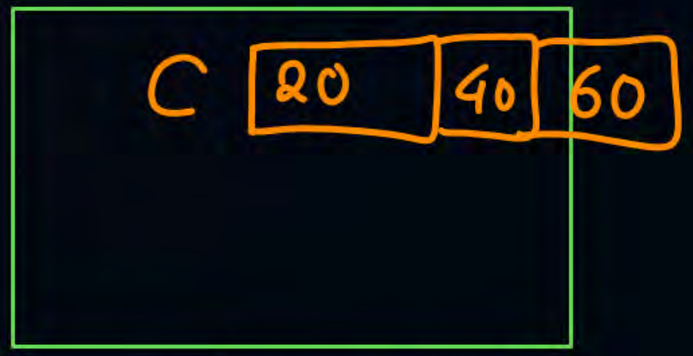
60

Slide



Recursion





Static data

Slide



Recursion

```
void f1() {
    static int a = 10;
    a = 20;
}
```

← once

← assignment ← Every time

Slide



2 mins Summary

Topic

Topic

Topic

Topic

Topic

Activation Record with stack (Run time)

Local variable (auto)

global variable (static)

Local can also be static

THANK - YOU

